

Curriculum Vitae

Personal Information

Name: Sung Yun Lee
Contact: sungyun98@g.postech.edu
Homepage: <https://sungyun98.github.io>
Languages: Korean (native), English



Skills

Experimental Techniques

- (Time-resolved) coherent diffraction imaging and ptychography
- (Resonant) X-ray diffraction and reciprocal-space mapping
- X-ray absorption spectroscopy

Software Development

- Algorithms, AI models, and data analysis in Python (PyTorch), MATLAB, and C/C++
- Device control with LabVIEW and EPICS (bluesky)
- Two-temperature model and solid-mechanics simulation in COMSOL Multiphysics

Device Development

- Design of diffractive and refractive optics for hard X-ray vortex beam generation
- (Time-resolved) reflective optical ptychography setup with LabVIEW and EPICS (bluesky)
- Cryogenic magneto-optical chamber with permanent-magnet system

Education

Ph.D. in Physics

03/2020–08/2025

POSTECH, Pohang 37673, KR

Thesis: *Development of functional coherent X-ray nanoimaging with deep-learning methods*

Advisor: Prof. Changyong Song

B.Sc. in Physics, Minor in Electrical Engineering – *magna cum laude*

03/2016–02/2020

POSTECH, Pohang 37673, KR

Thesis: *A Comprehensive Evaluation of the Process of Copying a Complex Figure in Early- and Late-Onset Alzheimer Disease*

Advisors: Dr. Jee Hyun Choi* and Prof. Changyong Song

(*Korea Institute of Science and Technology, Seoul 02792, KR)

Experience

Postdoctoral Research Associate 09/2026–present

Femtosecond X-ray Diffraction and Imaging Laboratory, Department of Physics, POSTECH, Pohang 37673, KR

- Topic: *Understanding topological properties of quantum materials based on vortex beam (continued)*

Postdoctoral Researcher (as technical research personnel) 09/2025–08/2026

Center for Ultrafast Science on Quantum Matter, Max Planck POSTECH/Korea Research Initiative, Pohang 37673, KR

- Topic: *Understanding topological properties of quantum materials based on vortex beam*

Undergraduate Researcher 03/2019–02/2020

Femtosecond X-ray Diffraction and Imaging Laboratory, Department of Physics, POSTECH, Pohang 37673, KR

- Topic: *Phase retrieval of chromosome diffraction patterns*

Intern 06/2019–08/2019

Machine Learning Team, Columbus Center, Netmarble Corp., Seoul 08379, KR

- Topic: *Analysis and detection of users cheating in MMORPG*

Research Trainee 06/2018–08/2018

Convergence Research Center for Diagnosis, Treatment, and Care System of Dementia, Brain Science Institute, Korea Institute of Science and Technology, Seoul 02792, KR

- Topic: *Analysis of data on dementia patients based on statistical physics*

Intern 06/2017–08/2017

PnA Team, Accelerator Business Division, Dawonsys Co., Ltd., Ansan 15655, KR

- Topic: *Investigation of target systems and moderators for boron neutron capture therapy*

Teaching Assistant

Department of Physics, POSTECH, Pohang 37673, KR

- PHYS199: Freshman Research Participation 09/2021–12/2021
- PHYS103: General Physics Laboratory I 03/2021–06/2021
- PHYS250: Physics Laboratory I 09/2020–12/2020

Mentor, Student Mentoring Program

POSTECH, Pohang 37673, KR

- CSED101: Programming and Problem Solving 09/2019–12/2019
- CSED101: Programming and Problem Solving 09/2018–12/2018
- PHYS101: General Physics I 03/2018–06/2018

Mentor, The 13th University Student Knowledge Volunteering KOSAF Camp 08/2016

Korea Student Aid Foundation, Daegu 41200, KR

- School: Jeonggok Middle School, Uiryeong 52125, KR
- Topic: Tessellation and the four-color theorem

Honors and Awards

Domestic Postdoctoral Fellowship	2026–2028
National Research Foundation of Korea	
Graduate Student Excellent Paper Award	2024
Korean Synchrotron Radiation User's Association	
BK21 POSTECHIAN Fellowship – Innovation	2024
POSTECH	
Graduate Student Excellent Paper Award	2023
Department of Physics, POSTECH	
Presidential Science Scholarship	2016–2019
Ministry of Science, ICT, and Future Planning	
Encouragement Prize, The 31st Humantech Paper Awards (coauthor)	2025
Samsung Electronics Co., Ltd.	
Best Poster Award	
The 35th Synchrotron Radiation Users' Workshop, Korean Synchrotron Radiation User's Association	2023
The 3rd PAL-XFEL Users' Meeting, Korean Synchrotron Radiation User's Association	2023
The 34th Synchrotron Radiation Users' Workshop, Korean Synchrotron Radiation User's Association	2022
Best Teaching Assistant Award	2020
Department of Physics, POSTECH	
BK21 First Paper Award	2020
Department of Physics, POSTECH	
Best Bachelor's Thesis Award	2019
Department of Physics, POSTECH	

Patents

[1]	Apparatus and Method of Processing Image Copying Test for Evaluating Cognitive Impairment, and Computer Readable Recording Medium	
	Registration #10-2338071, KR	12/09/2021
[2]	METHOD FOR GAME DATA PROCESSING	
	Registration #10-2333941, KR	11/29/2021

Publications (†: co-first author, *: corresponding author)

- [1] Sung Yun Lee, Seong Gook Kim, Heung-Sik Park, Jihun Kim, Jungchan Choi, Sinwoo Kim, Eunyoun Park, Seung-Phil Heo, Seongbin Oh, Junha Hwang, Myong-jin Kim, Minhyun Kim, Do Geun Jang, Hyunjung Kim, Daewoong Nam, Sangsoo Kim, Hyon Chol Kang, Chan-Ho Yang, Changyong Song*, *Nanolayer-Confined Strain Waves in Femtosecond-Optical-Vortex-Excited Multiferroic BiFeO₃*, Nano Letters, accepted (2026).
- [2] Jaeyong Shin, Junha Hwang, Sejin Kim, Sung Yun Lee, Eunyoun Park, Sangsoo Kim, Changyong Song, and Daewoong Nam*, *Improved image reconstruction in coherent diffraction imaging using self-seeded XFEL pulses*, Journal of Synchrotron Radiation **33**(4), 904–912 (2026).
- [3] Eunyoun Park†, Chulho Jung†, Junha Hwang, Jaeyong Shin, Sung Yun Lee, Heemin Lee, Seung-Phil Heo, Daewoong Nam, Sangsoo Kim, Min Seok Kim, Kyung Sook Kim, In Tae Eom, Do Young Noh, Yungok Ihm, and Changyong Song*, *Surface-plasmon control of ultrafast energy-relaxation modes in photoexcited Au nanorods probed by time-resolved single-particle X-ray imaging*, Nature Communications **16**, 9876 (2025).
- [4] Seung-Phil Heo, Choongjae Won, Heemin Lee, Hanbyul Kim, Eunyoun Park, Sung Yun Lee, Junha Hwang, Hyeonggi Choi, Sang-Youn Park, Byungjune Lee, Woo-Suk Noh, Hoyoung Jang, Jae-Hoon Park, Dongbin Shin*, and Changyong Song*, *Frustrated phonon with charge density wave in vanadium Kagome metal*, Nature Communications **16**, 4861 (2025).
- [5] Sung Yun Lee, Do Hyung Cho, Chulho Jung, Daeho Sung, Daewoong Nam, Sangsoo Kim, and Changyong Song*, *Deep-learning real-time phase retrieval of imperfect diffraction patterns from X-ray free-electron lasers*, npj Computational Materials **11**, 68 (2025).
- [6] Jangwoo Kim†, HyoJung Hyun†, Seonghan Kim, Sun Min Hwang, Myong-Jin Kim, Dogeun Jang, Kyung Sook Kim, Jaeyong Shin, Sejin Kim, Junha Hwang, Sung Yun Lee, Eunyoun Park, Sangsoo Kim, Intae Eom, Changyong Song, and Daewoong Nam*, *Development of the Nanobeam X-ray Experiments instrument at PAL-XFEL*, Journal of Synchrotron Radiation **32**(2), 466–473 (2025).
- [7] Sung Yun Lee, Eunyoun Park, Sinwoo Kim, Euije Jo, Su Yong Lee, Jun Woo Choi, and Changyong Song*, *Off-Axis X-Ray Vortex Beam Ptychography*, ACS Photonics **11**(9), 3804–3810 (2024).
- [8] Junha Hwang†, Yungok Ihm†, Daewoong Nam, Jaeyong Shin, Eunyoun Park, Sung Yun Lee, Heemin Lee, Seung-Phil Heo, Sangsoo Kim, Je Young Ahn, Ji Hoon Shim, Minseok Kim, Intae Eom, Do Young Noh, and Changyong Song*, *Inverted nucleation for photoinduced nonequilibrium melting*, Science Advances **10**(18), ead16409 (2024).
- [9] Junha Hwang, Sejin Kim, Sung Yun Lee, Eunyoun Park, Jaeyong Shin, Jae Hyuk Lee, Myong-jin Kim, Seonghan Kim, Sang-Youn Park, Dogeun Jang, Intae Eom, Sangsoo Kim, Changyong Song, Kyung Sook Kim*, and Daewoong Nam*, *Development of the multiplex imaging chamber at PAL-XFEL*, Journal of Synchrotron Radiation **31**(3), 469–477 (2024).
- [10] Sung Yun Lee†, Do Hyung Cho†, Sung Chan Song†, Jaeyong Shin, Junha Hwang, Eunyoun Park, Su Yong Lee, Seongseop Kim, Jinwoo Lee, and Changyong Song*, *Nanoscale Three-Dimensional Network Structure of a Mesoporous Particle Unveiled via Adaptive Multidistance Coherent X-ray Tomography*, ACS Nano **17**(22), 22488–22498 (2023).
- [11] Sung Yun Lee, Do Hyung Cho, Chulho Jung, Daeho Sung, Daewoong Nam, Sangsoo Kim, and Changyong Song*, *Denoising low-intensity diffraction signals using k-space deep learning: Applications to phase recovery*, Physical Review Research **3**(4), 043066 (2021).
- [12] Do Hyung Cho†, Zhou Shen†, Yungok Ihm, Dae Han Wi, Chulho Jung, Daewoong Nam, Sangsoo Kim,

Sang-Youn Park, Kyung Sook Kim, Daeho Sung, Heemin Lee, Jae-Yong Shin, Junha Hwang, Sung Yun Lee, Su Yong Lee, Sang Woo Han, Do Young Noh, N. Duane Loh*, and Changyong Song*, *High-Throughput 3D Ensemble Characterization of Individual Core-Shell Nanoparticles with X-ray Free Electron Laser Single-Particle Imaging*, ACS Nano **15**(3), 4066–4076 (2021).

- [13] Ko Woon Kim[†], Sung Yun Lee[†], Jongdoo Choi, Juhee Chin, Byung Hwa Lee, Duk L. Na*, and Jee Hyun Choi*, *A Comprehensive Evaluation of the Process of Copying a Complex Figure in Early- and Late-Onset Alzheimer Disease: A Quantitative Analysis of Digital Pen Data*, Journal of Medical Internet Research **22**(8), e18136 (2020).
- [14] Sungyun Lee*, Sunghun Kim, Inhae Seok, and Mincheol Kim, *Detecting Abuser Group in MMORPG by using Ranking System based on Game Transaction Network*, In Proceedings of the Korea Software Congress, 584–586 (KIISE, 2019).

Presentations

- [1] The 27th Congress of the International Union of Crystallography (Oral, as an invited talk) 2026
International Union of Crystallography
Title: *Deep-learning real-time phase retrieval of imperfect diffraction patterns from X-ray free-electron lasers*
- [2] The 6th PAL-XFEL Users' Meeting (Oral, as an invited talk) 2026
Korean Synchrotron Radiation User's Association
Title: *Nanolayer-Confined Strain Waves in Femtosecond-Optical-Vortex-Excited Multiferroic BiFeO₃*
- [3] The 37th Synchrotron Radiation Users' Workshop (Oral, as an invited talk) 2025
Korean Synchrotron Radiation User's Association
Title: *Deep learning and phase retrieval for coherent diffraction imaging*
- [4] The 5th PAL-XFEL Users' Meeting (Oral, as an invited student talk) 2025
Korean Synchrotron Radiation User's Association
Title: *Deep learning for coherent diffraction imaging using XFELs*
- [5] The 36th Synchrotron Radiation Users' Workshop (Poster) 2024
Korean Synchrotron Radiation User's Association
Title: *Off-axis X-ray vortex beam ptychography*
- [6] APS March Meeting 2024 (Oral) 2024
American Physical Society
Title: *Nanoscale X-ray Tomography of Mesoporous Particle Improved via Adaptive Multidistance Coherent Diffraction Imaging*
- [7] The 35th Synchrotron Radiation Users' Workshop (Poster) 2023
Korean Synchrotron Radiation User's Association
Title: *Network analysis of pore structure inside mesoporous particle revealed by multi-distance coherent X-ray tomography*
- [8] 2023 KPS Fall Meeting (Poster) 2023
Korean Physical Society
Title: *Nanoscale three-dimensional network structure of a mesoporous particle unveiled via adaptive multi-distance coherent X-ray tomography*
- [9] The 3rd PAL-XFEL Users' Meeting (Poster) 2023
Korean Synchrotron Radiation User's Association

Title: *Deep-Learning-Based Denoiser for Phase Recovery of Single-Shot Diffraction Signals via X-ray Free Electron Laser*

- [10] The 34th Synchrotron Radiation Users' Workshop (Poster) 2022
Korean Synchrotron Radiation User's Association
Title: *Multi-distance Coherent Diffraction Imaging for Super-resolution X-ray Microscopy*
- [11] 2021 KPS Fall Meeting (Poster) 2021
Korean Physical Society
Title: *Deep-Learning-Based Denoiser for Phase Recovery of Single-Shot Diffraction Patterns Using XFEL*
- [12] The 32nd Synchrotron Radiation Users' Workshop (Poster) 2020
Korean Synchrotron Radiation User's Association
Title: *Low Intensity Phase Retrieval Enhanced by Deep Neural Network*
- [13] Korea Software Congress 2019 (Poster) 2019
Korean Institute of Information Scientists and Engineers
Title: *Detecting Abuser Group in MMORPG by using Ranking System based on Game Transaction Network*